Column Name	name_length
pelvic_incidence	16
pelvic_tilt	11
lumbar_lordosis	15
sacral_slope	12
pelvic_radius	13
spondylolisthesis_grade	23
Class_att	9

The TTEST Procedure

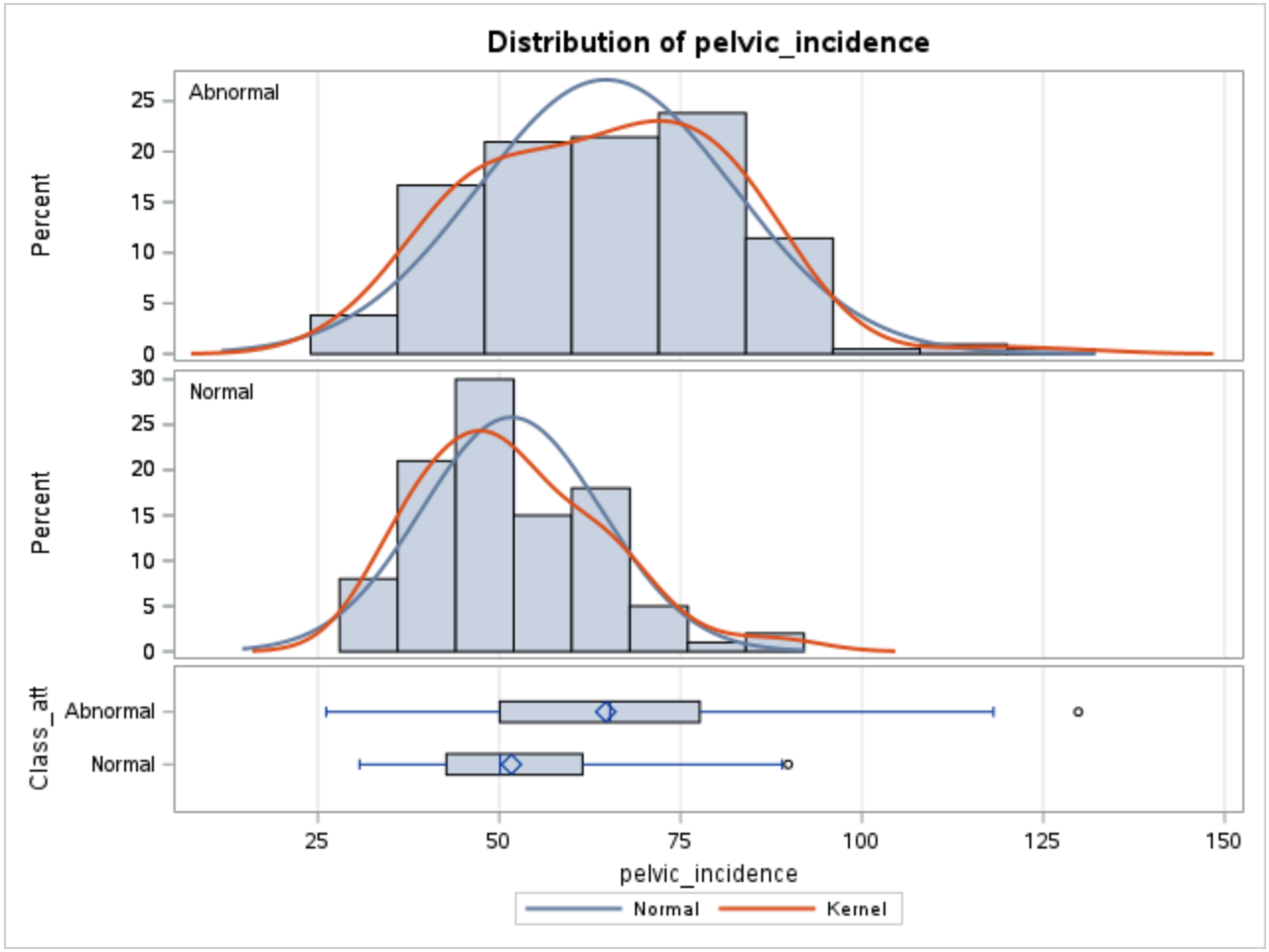
Variable: pelvic_incidence

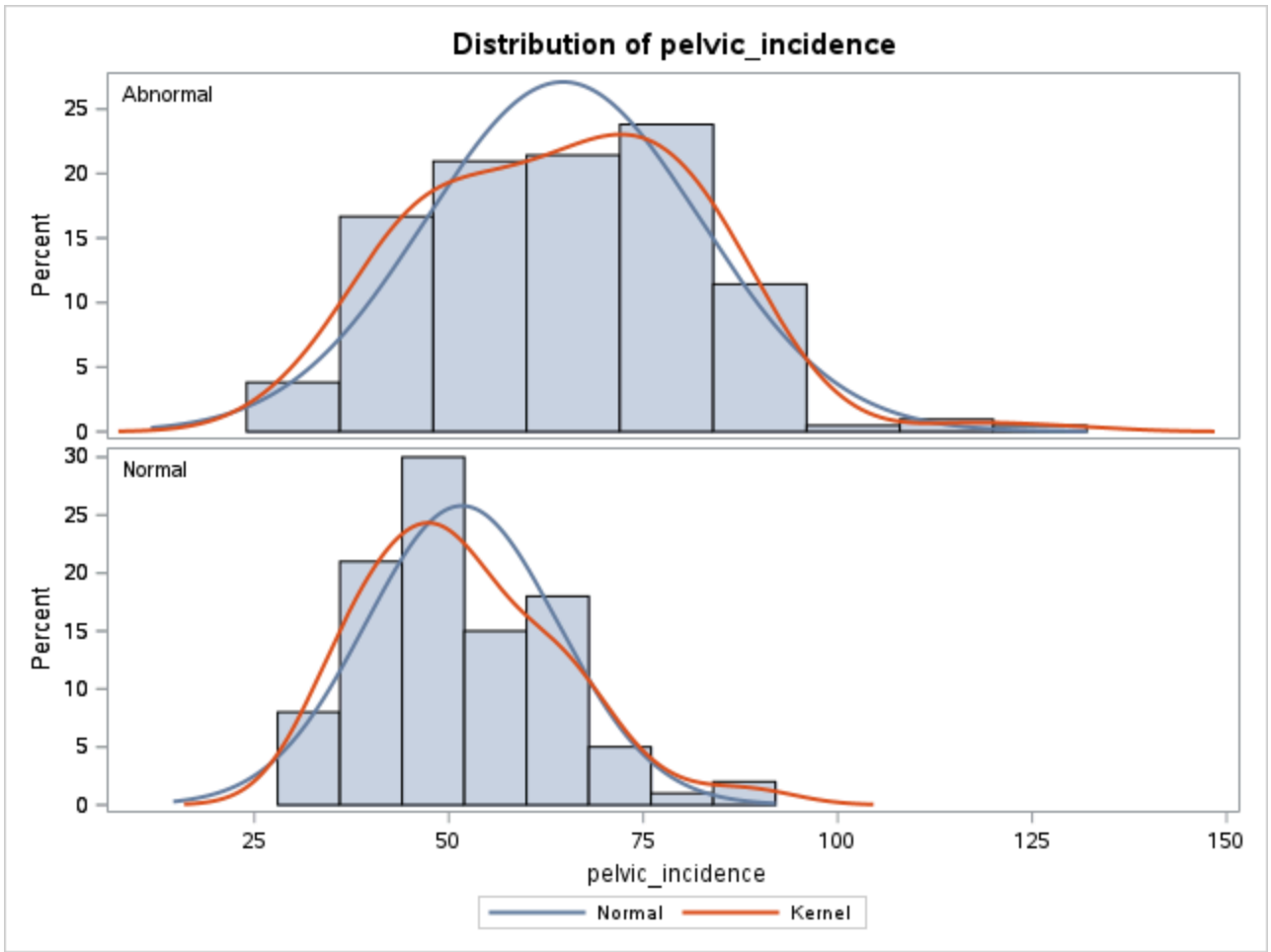
Class_att	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Abnormal		210	64.6926	17.6621	1.2188	26.1479	129.8
Normal		100	51.6852	12.3682	1.2368	30.7419	89.8347
Diff (1-2)	Pooled		13.0073	16.1509	1.9623		
Diff (1-2)	Satterthwaite		13.0073		1.7364		

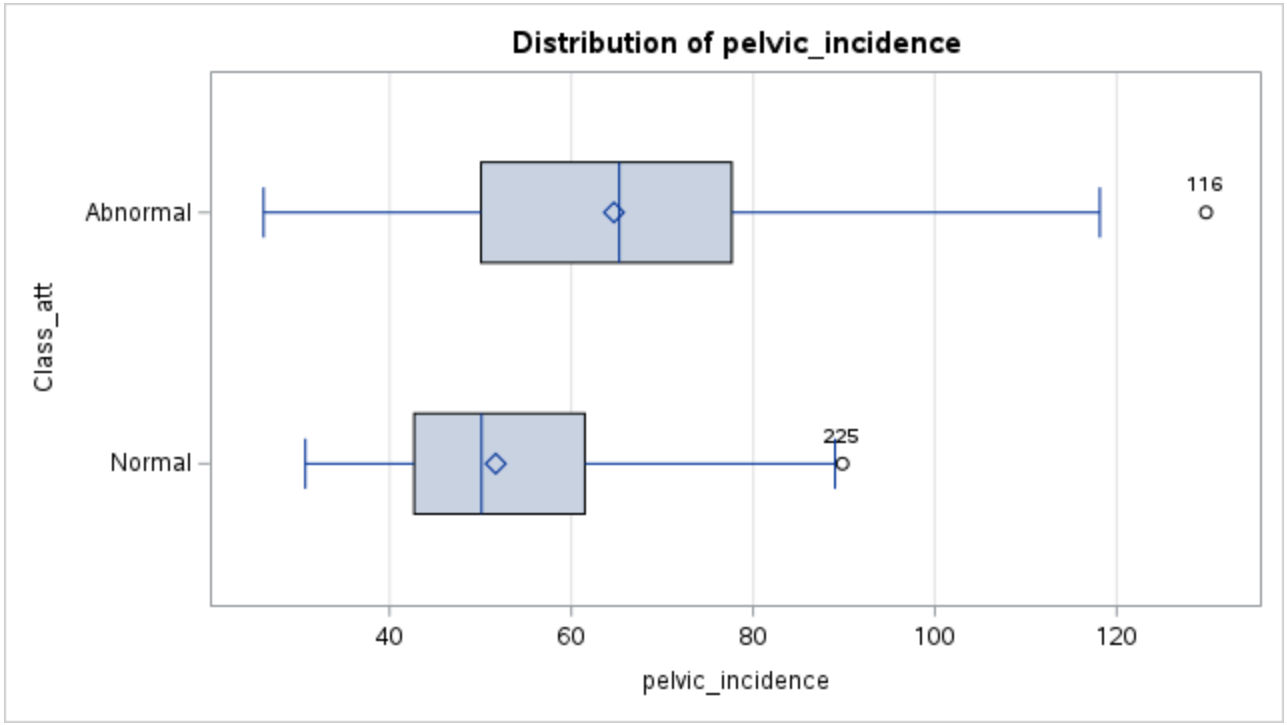
Class_att	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Abnormal		64.6926	62.2898	67.0953	17.6621	16.1190	19.5345
Normal		51.6852	49.2311	54.1394	12.3682	10.8593	14.3678
Diff (1-2)	Pooled	13.0073	9.1461	16.8685	16.1509	14.9699	17.5356
Diff (1-2)	Satterthwaite	13.0073	9.5884	16.4262			

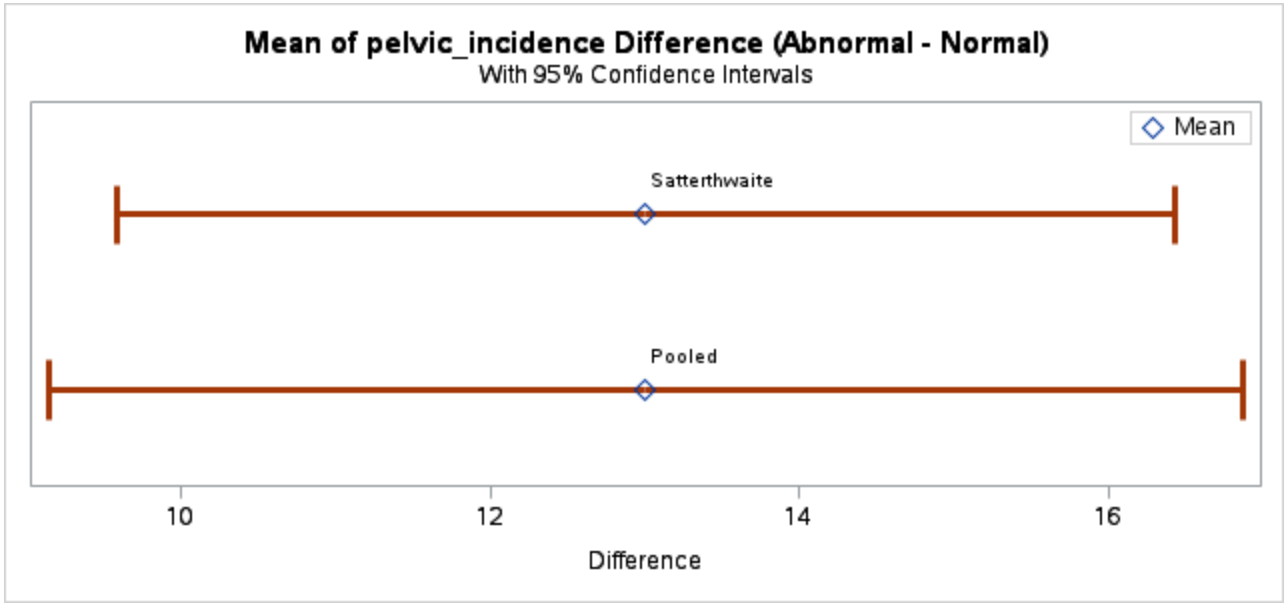
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	308	6.63	<.0001
Satterthwaite	Unequal	265.87	7.49	<.0001

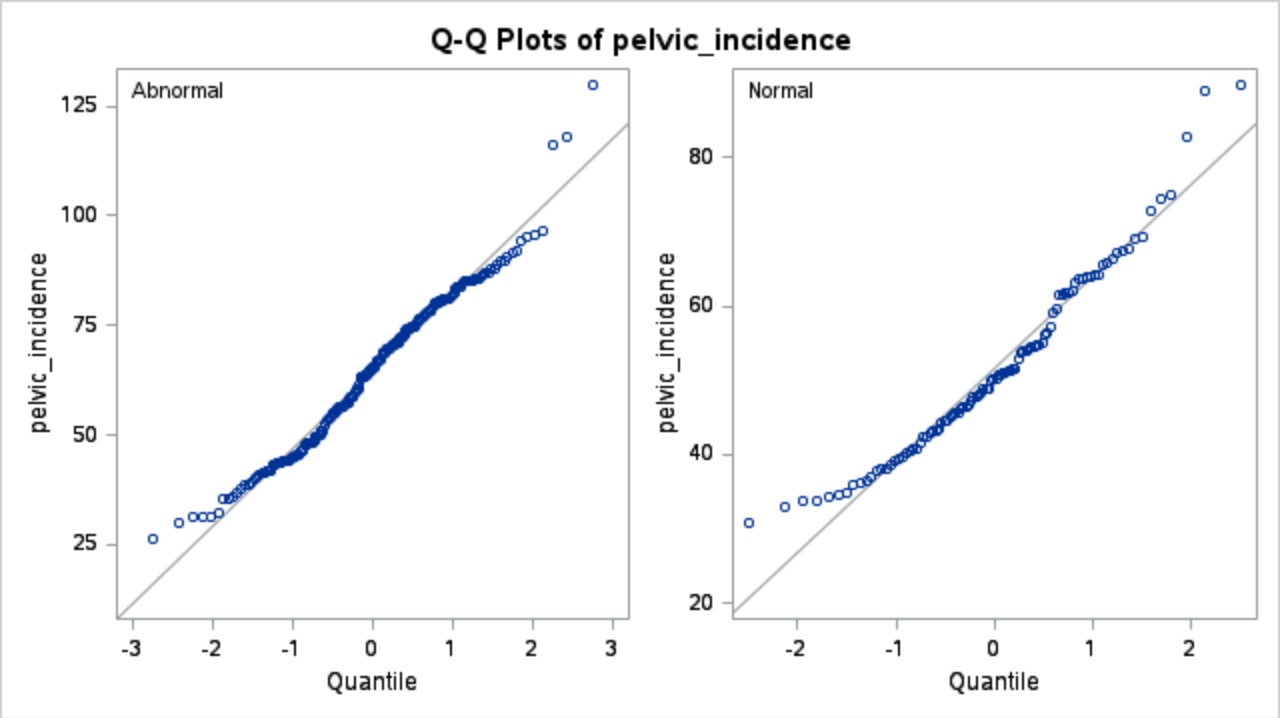
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	209	99	2.04	<.0001


Distribution of pelvic_incidence
Abnormal
Normal
Percent
Class_att
Abnormal
Normal
pelvic_incidence
Normal
Kernel


Distribution of pelvic_incidence
Abnormal
Normal
Percent
pelvic_incidence
Normal
Kernel


Distribution of pelvic_incidence
Abnormal
Normal
Class_att
116
225
40
60
80
100
120
pelvic_incidence


Mean of pelvic_incidence Difference (Abnormal - Normal)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
10
12
14
16
Difference



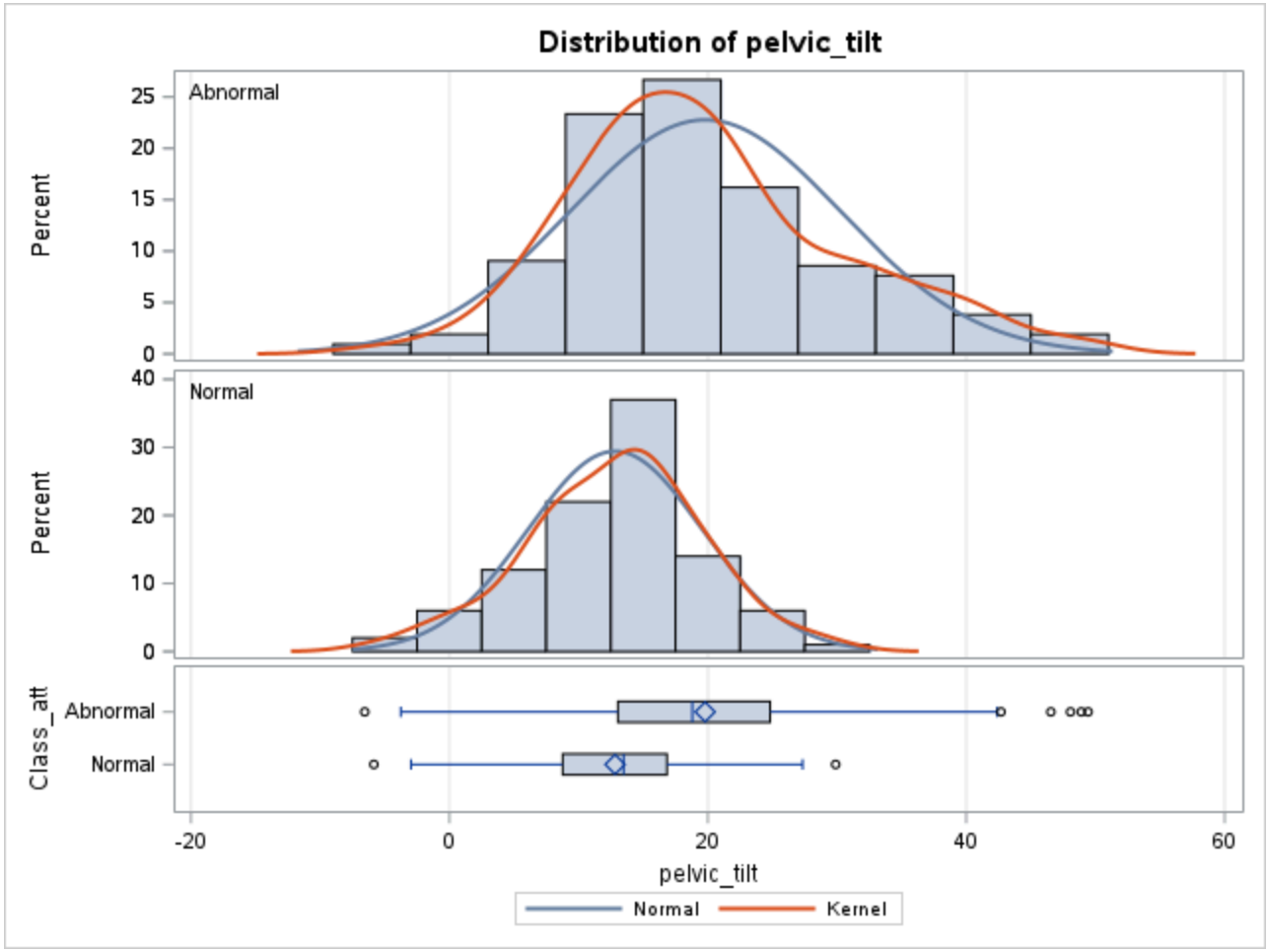
Variable: pelvic_tilt

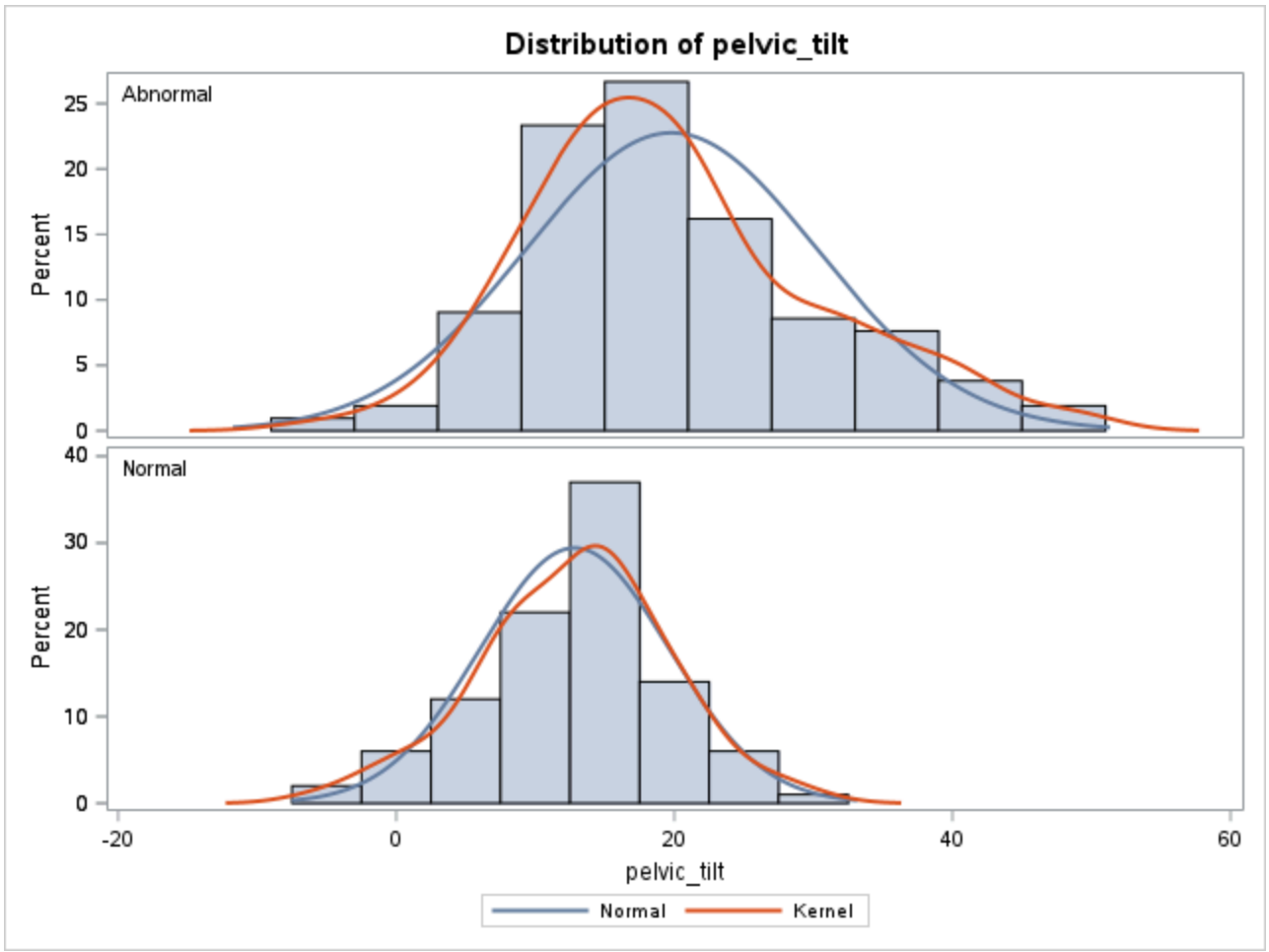
Class_att	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Abnormal		210	19.7911	10.5159	0.7257	-6.5549	49.4319
Normal		100	12.8214	6.7785	0.6779	-5.8460	29.8941
Diff (1-2)	Pooled		6.9697	9.4767	1.1514		
Diff (1-2)	Satterthwaite		6.9697		0.9930		

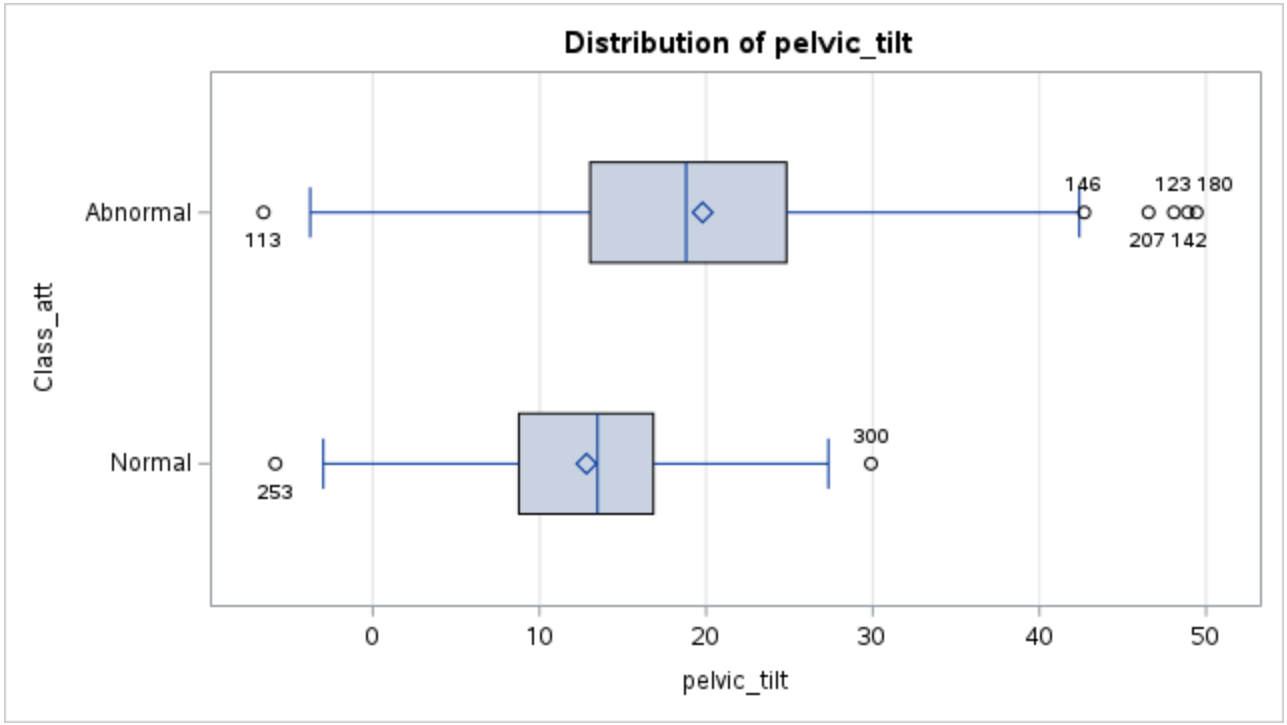
Class_att	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Abnormal		19.7911	18.3606	21.2217	10.5159	9.5971	11.6307
Normal		12.8214	11.4764	14.1664	6.7785	5.9516	7.8744
Diff (1-2)	Pooled	6.9697	4.7041	9.2353	9.4767	8.7838	10.2892
Diff (1-2)	Satterthwaite	6.9697	5.0150	8.9244			

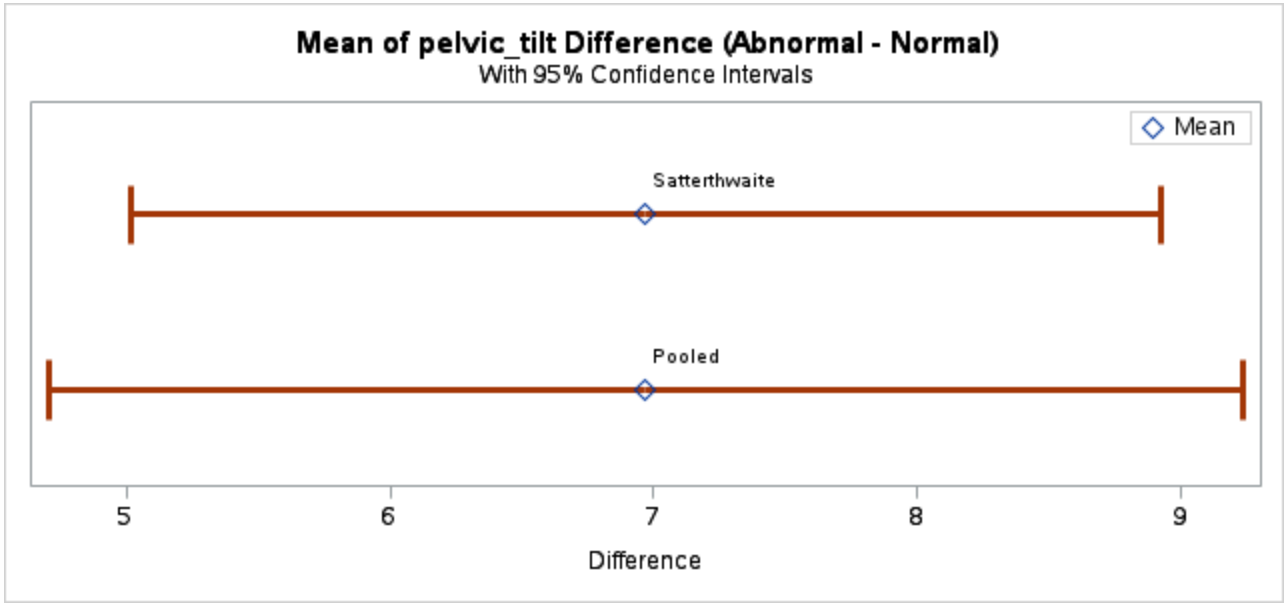
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	308	6.05	<.0001
Satterthwaite	Unequal	281.08	7.02	<.0001

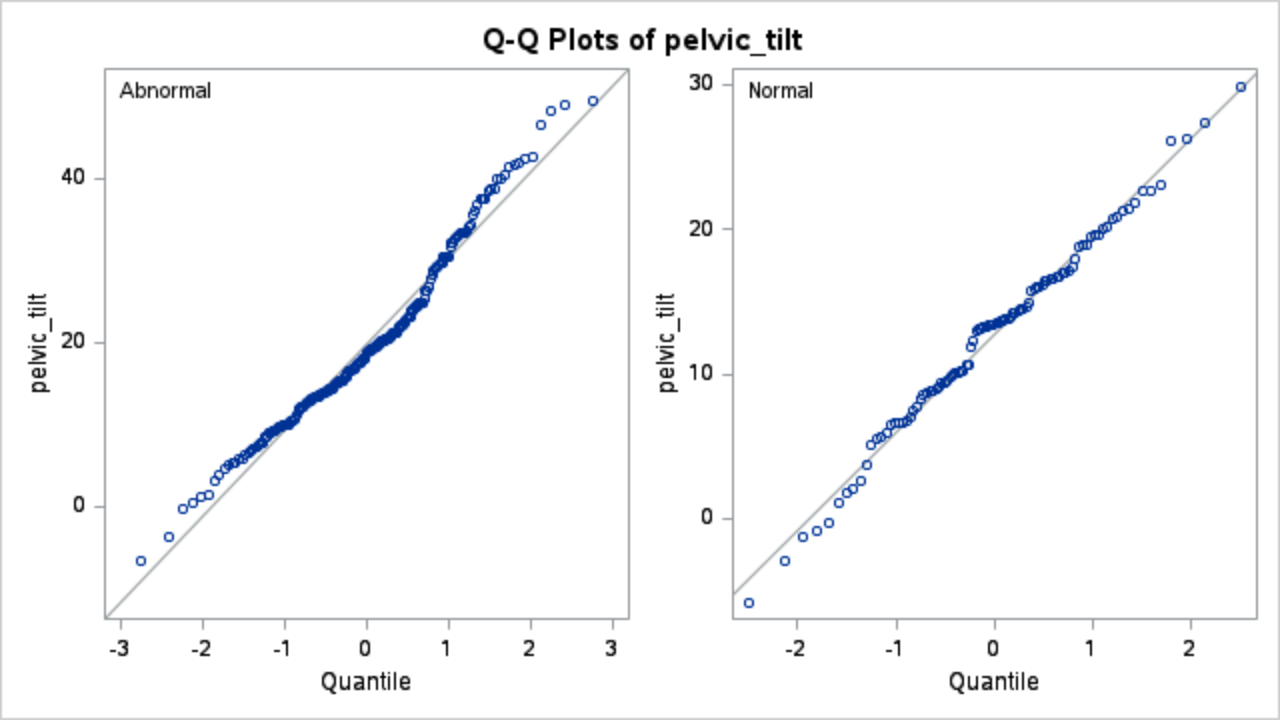
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	209	99	2.41	<.0001


Distribution of pelvic_tilt
Abnormal
Normal
Percent
Class_att
Abnormal
Normal
-20
0
20
40
60
pelvic_tilt
Normal
Kernel


Distribution of pelvic_tilt
Abnormal
Normal
Percent
-20
0
20
40
60
pelvic_tilt
Normal
Kernel


Distribution of pelvic_tilt
Class_att
Abnormal
Normal
113
146
123 180
207 142
253
300
0
10
20
30
40
50
pelvic_tilt


Mean of pelvic_tilt Difference (Abnormal - Normal)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
5
6
7
8
9
Difference



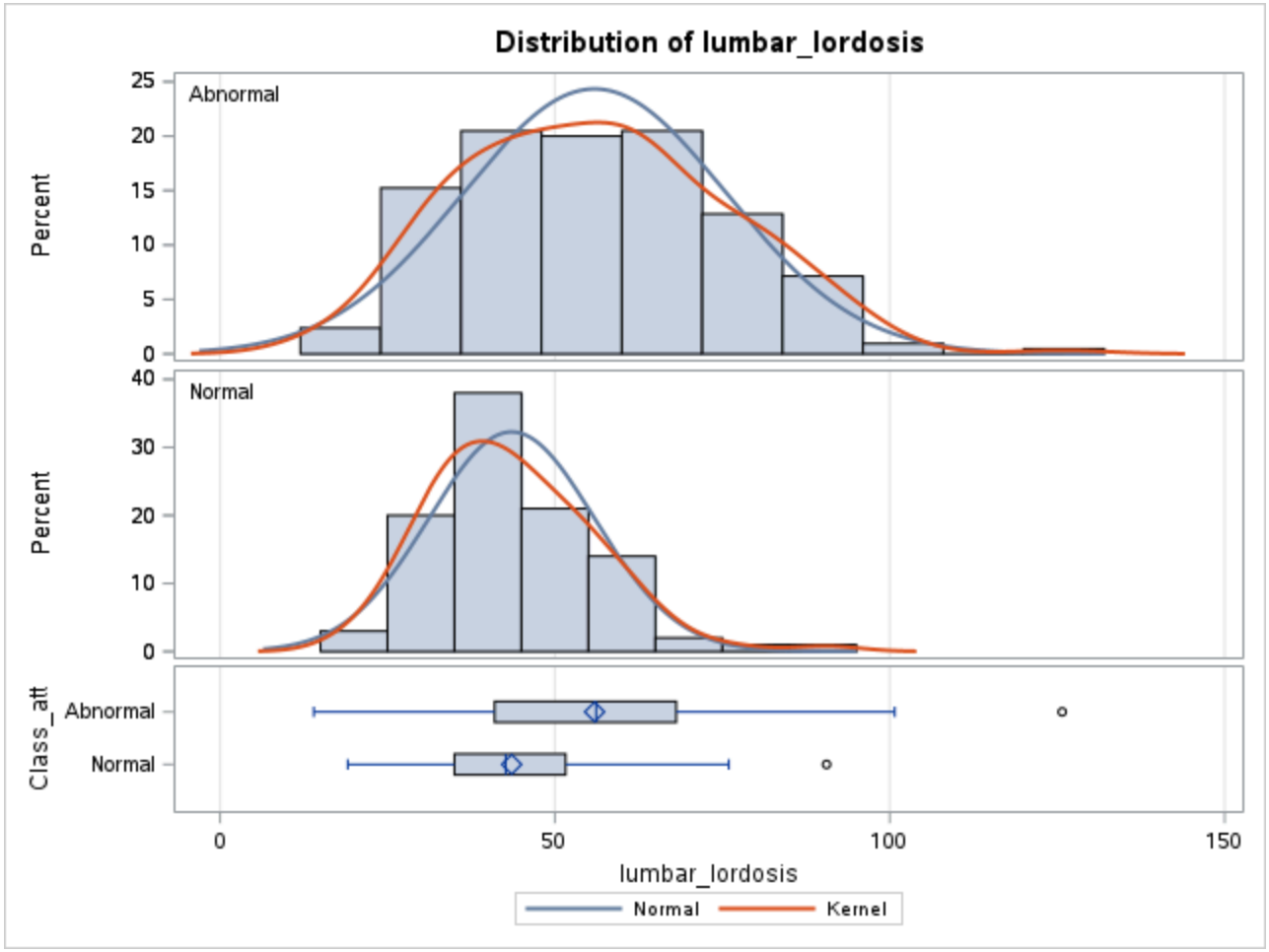
Variable: lumbar_lordosis

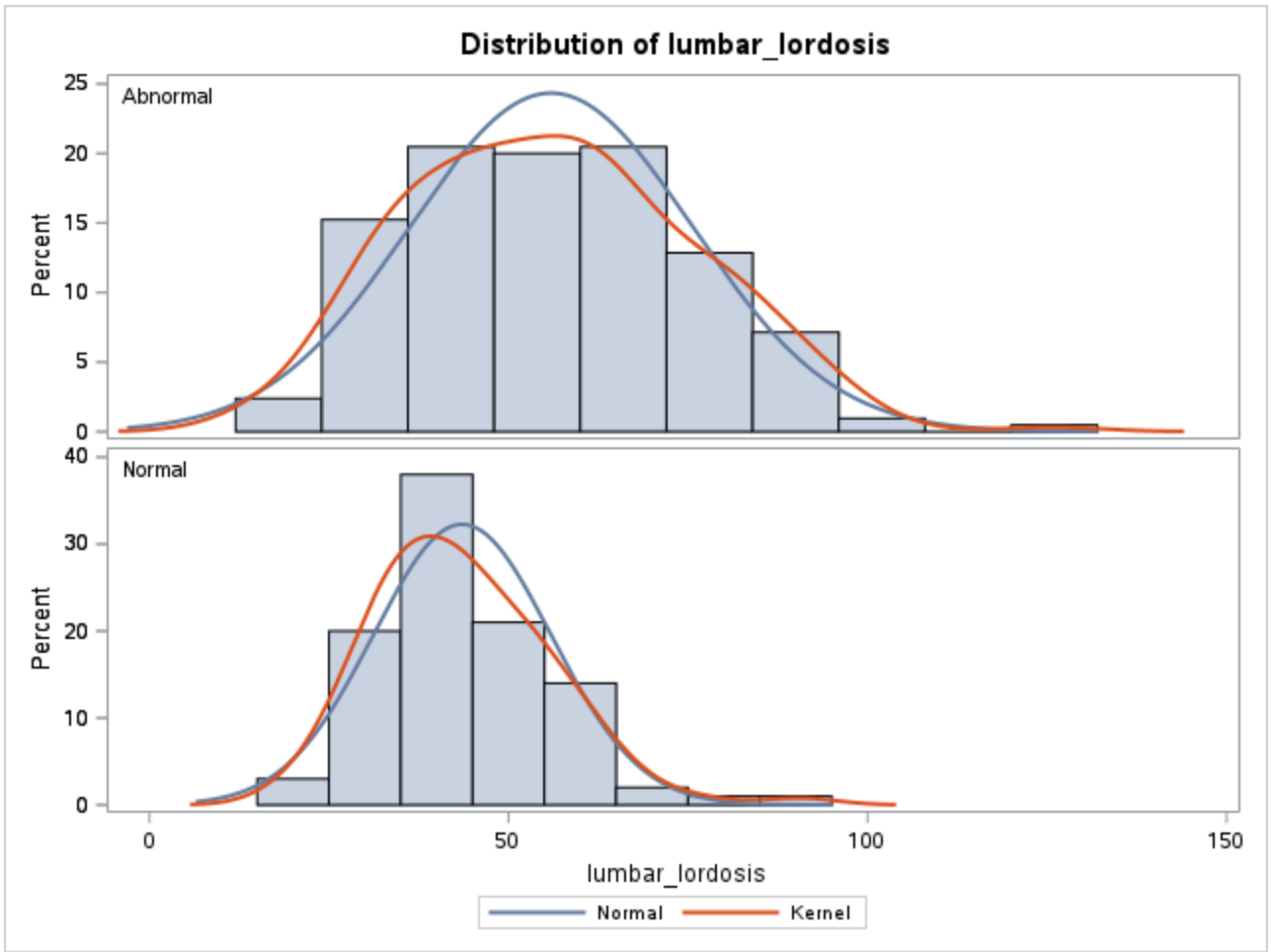
Class_att	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Abnormal		210	55.9254	19.6695	1.3573	14.0000	125.7
Normal		100	43.5426	12.3614	1.2361	19.0711	90.5635
Diff (1-2)	Pooled		12.3828	17.6535	2.1449		
Diff (1-2)	Satterthwaite		12.3828		1.8359		

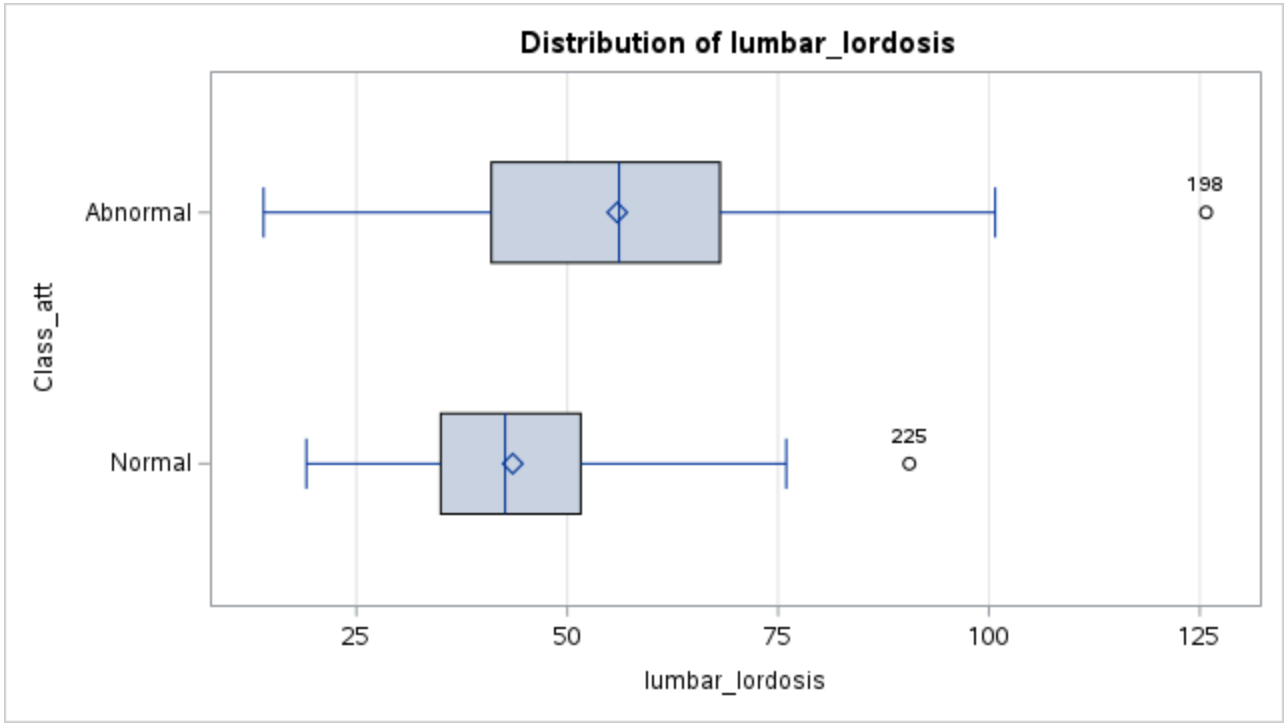
Class_att	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Abnormal		55.9254	53.2496	58.6012	19.6695	17.9510	21.7546
Normal		43.5426	41.0898	45.9954	12.3614	10.8534	14.3599
Diff (1-2)	Pooled	12.3828	8.1623	16.6032	17.6535	16.3627	19.1671
Diff (1-2)	Satterthwaite	12.3828	8.7692	15.9963			

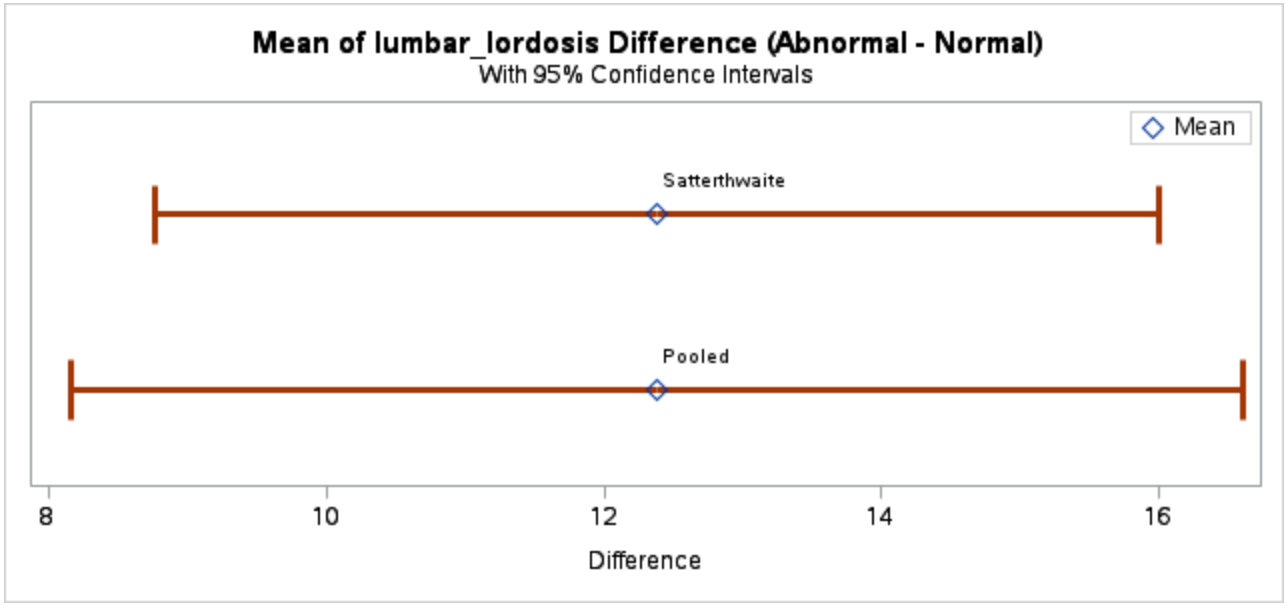
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	308	5.77	<.0001
Satterthwaite	Unequal	285.23	6.74	<.0001

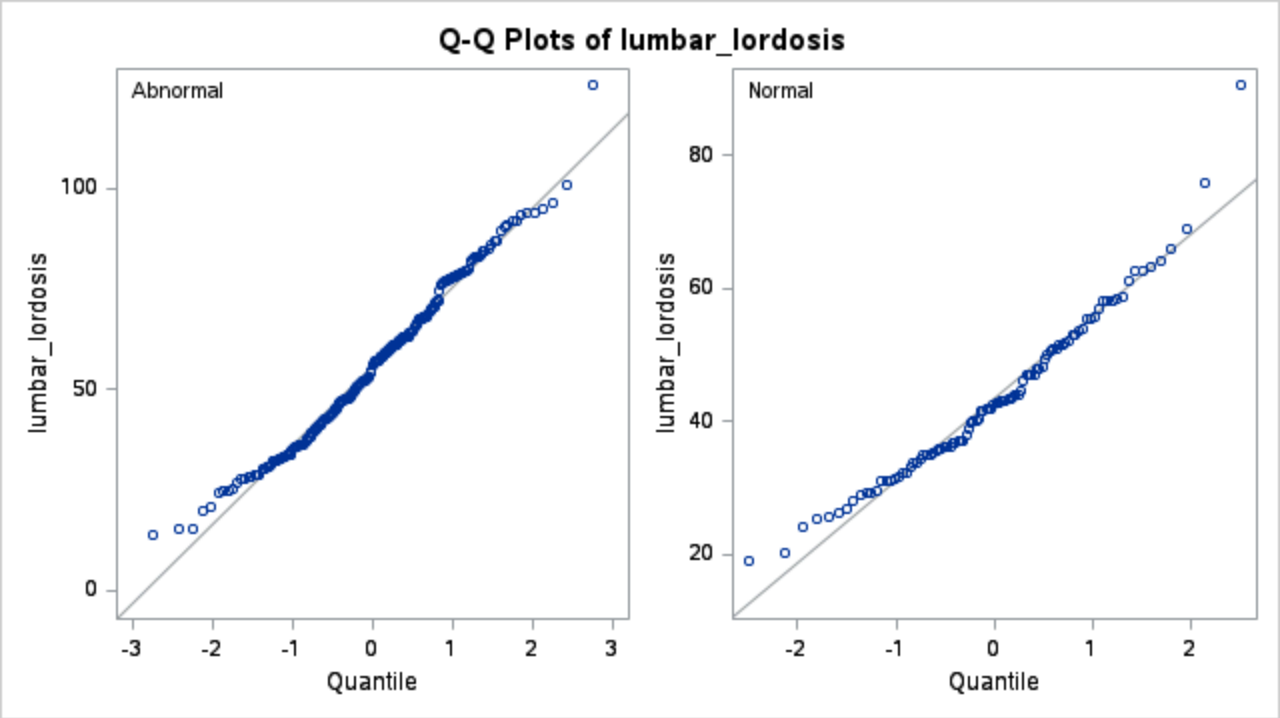
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	209	99	2.53	<.0001


Distribution of lumbar_lordosis
Abnormal
Normal
Percent
Class_att
Abnormal
Normal
0
50
100
150
lumbar_lordosis
Normal
Kernel


Distribution of lumbar_lordosis
Abnormal
Normal
Percent
0
50
100
150
lumbar_lordosis
Normal
Kernel


Distribution of lumbar_lordosis
Abnormal
Normal
Class_att
198
225
25
50
75
100
125
lumbar_lordosis


Mean of lumbar_lordosis Difference (Abnormal - Normal)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
8
10
12
14
16
Difference



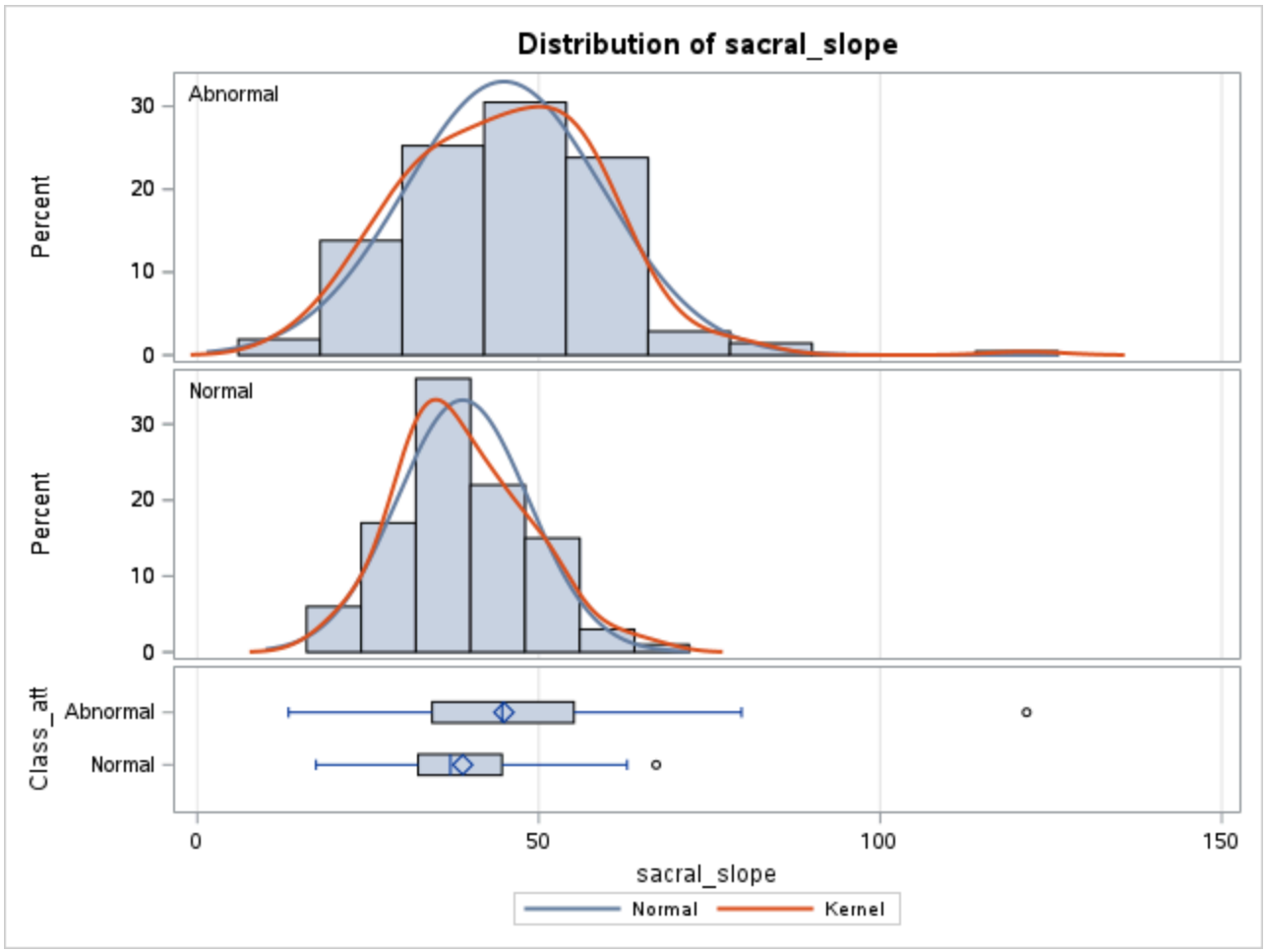
Variable: sacral_slope

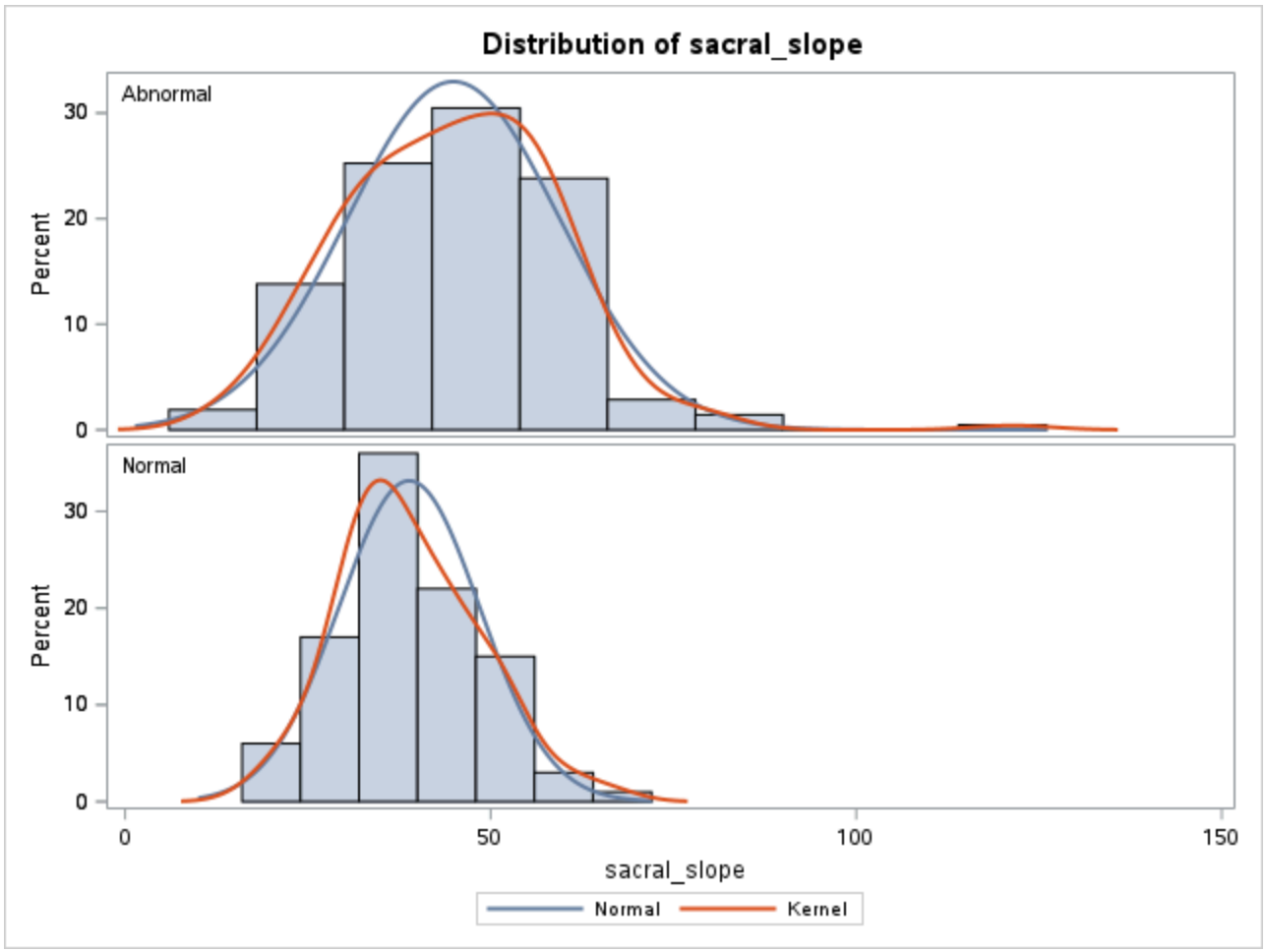
Class_att	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Abnormal		210	44.9015	14.5156	1.0017	13.3669	121.4
Normal		100	38.8638	9.6240	0.9624	17.3870	67.1955
Diff (1-2)	Pooled		6.0376	13.1433	1.5969		
Diff (1-2)	Satterthwaite		6.0376		1.3891		

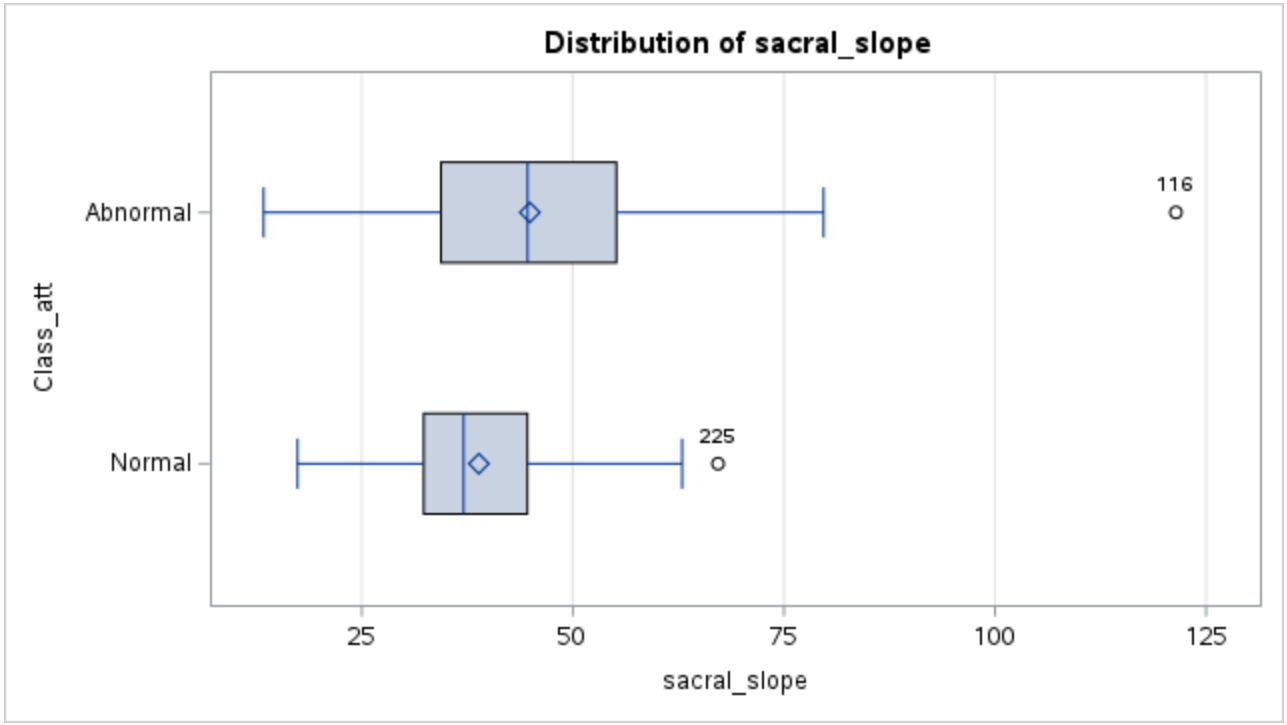
Class_att	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Abnormal		44.9015	42.9268	46.8761	14.5156	13.2474	16.0544
Normal		38.8638	36.9542	40.7734	9.6240	8.4499	11.1800
Diff (1-2)	Pooled	6.0376	2.8954	9.1798	13.1433	12.1823	14.2702
Diff (1-2)	Satterthwaite	6.0376	3.3031	8.7722			

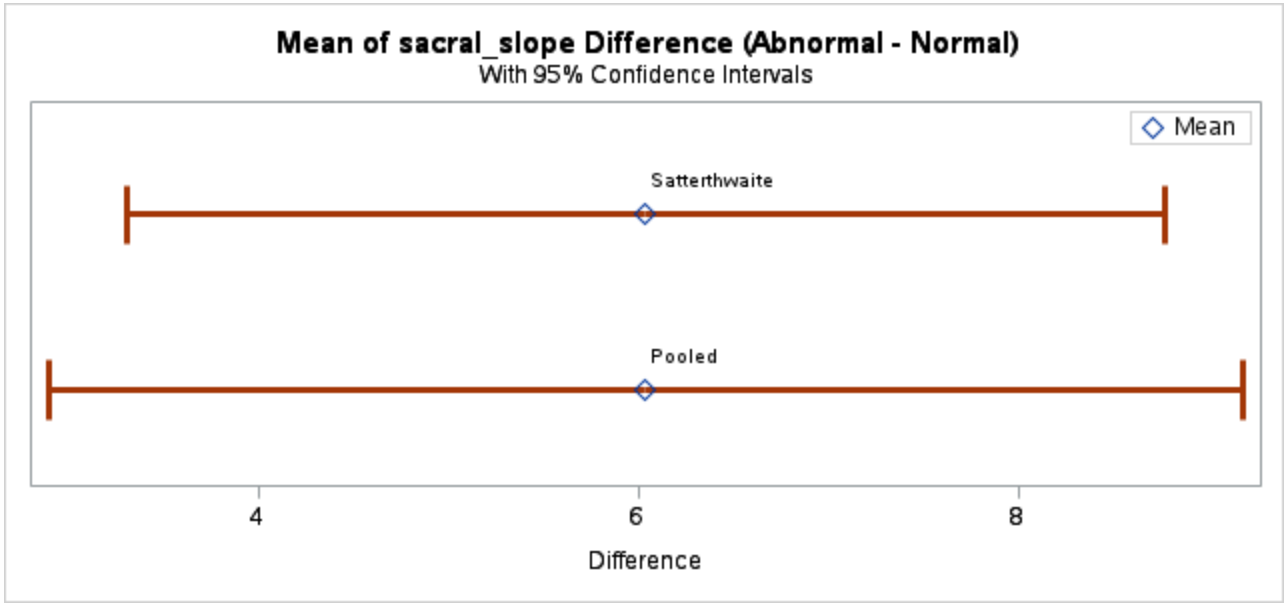
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	308	3.78	0.0002
Satterthwaite	Unequal	276.16	4.35	<.0001

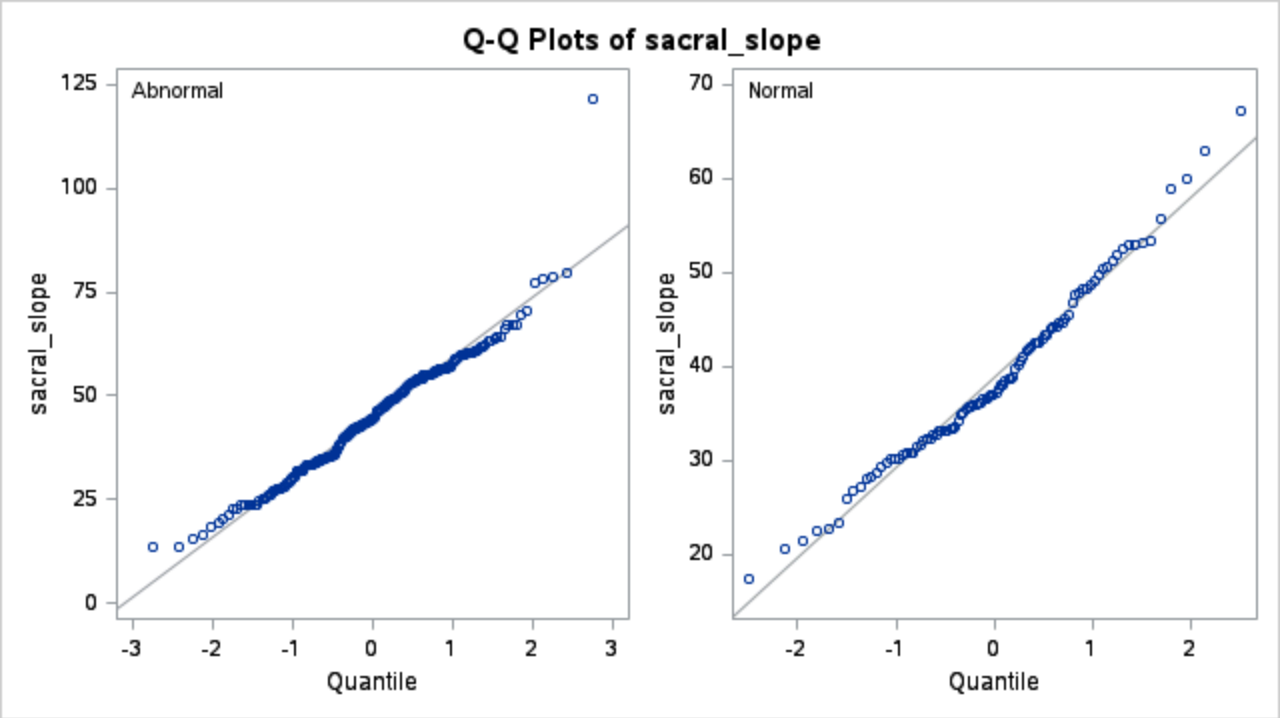
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	209	99	2.27	<.0001


Distribution of sacral_slope
Abnormal
Normal
Percent
Class_att
Abnormal
Normal
0
50
100
150
sacral_slope
Normal
Kernel


Distribution of sacral_slope
Abnormal
Normal
Percent
0
10
20
30
0
50
100
150
sacral_slope
Normal
Kernel


Distribution of sacral_slope
Abnormal
Normal
Class_att
116
225
25
50
75
100
125
sacral_slope


Mean of sacral_slope Difference (Abnormal - Normal)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
4
6
8
Difference



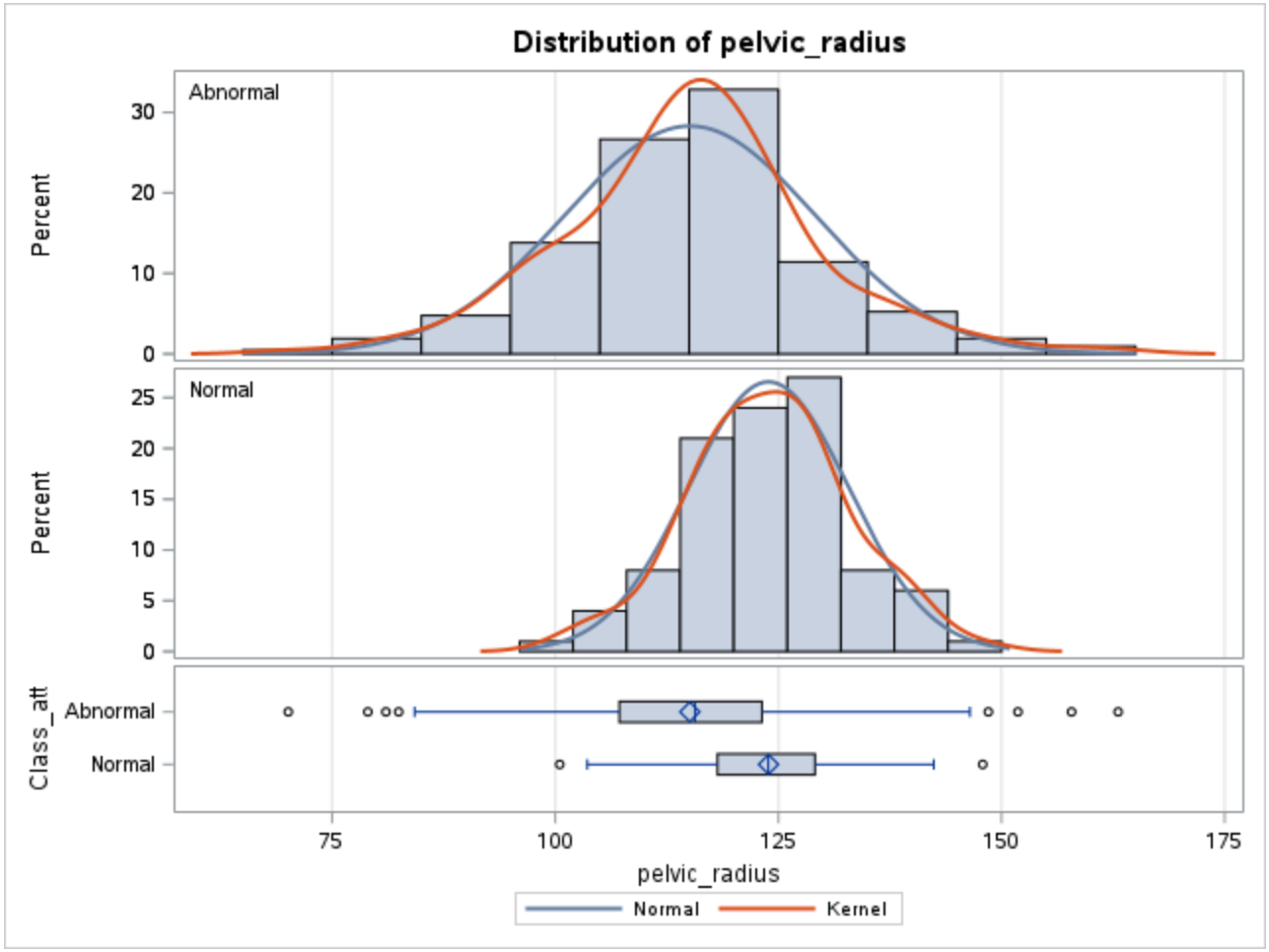
Variable: pelvic_radius

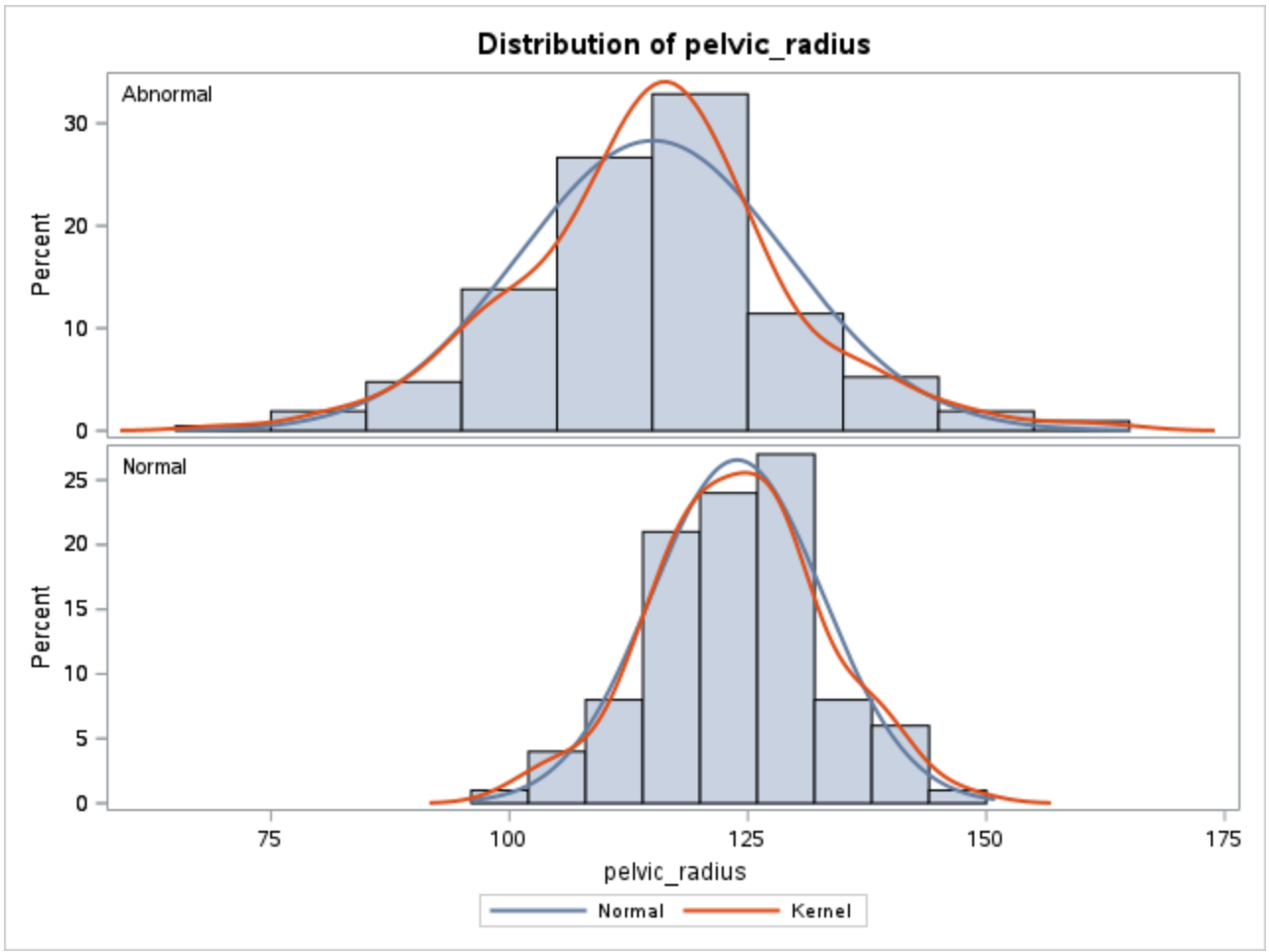
Class_att	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Abnormal		210	115.1	14.0906	0.9723	70.0826	163.1
Normal		100	123.9	9.0142	0.9014	100.5	147.9
Diff (1-2)	Pooled		-8.8131	12.6825	1.5409		
Diff (1-2)	Satterthwaite		-8.8131		1.3259		

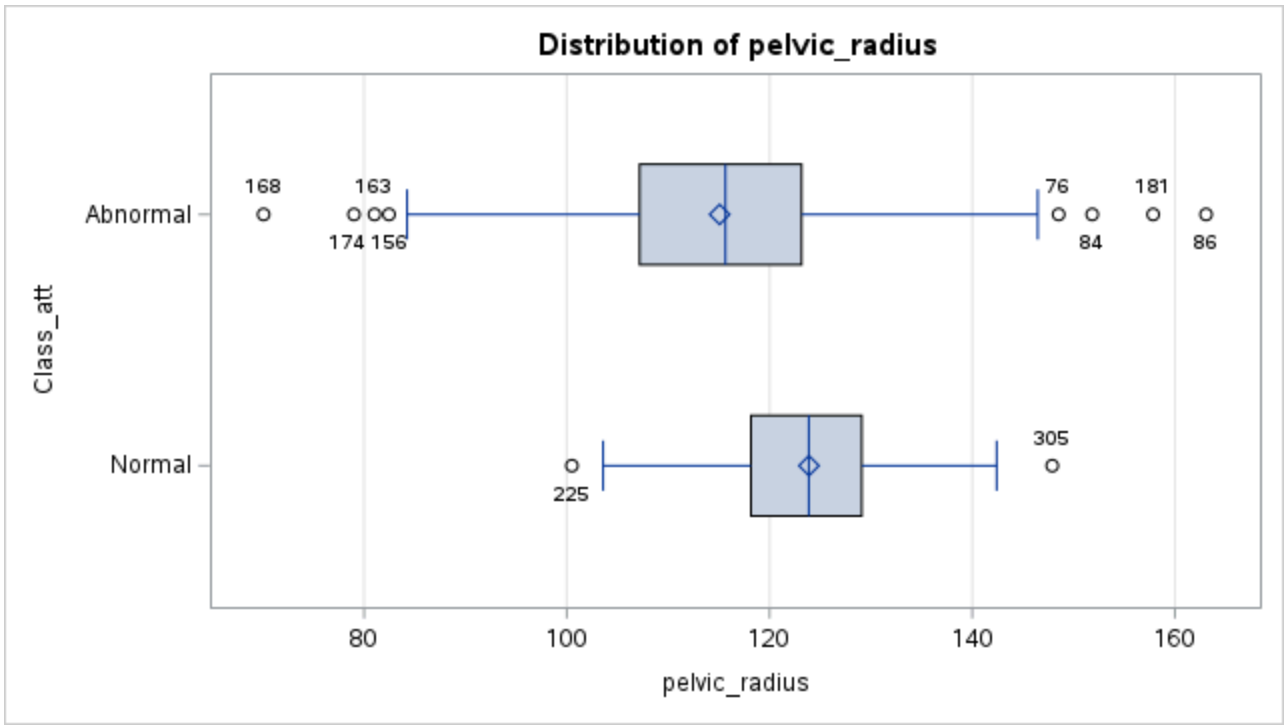
Class_att	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Abnormal		115.1	113.2	117.0	14.0906	12.8595	15.5844
Normal		123.9	122.1	125.7	9.0142	7.9146	10.4716
Diff (1-2)	Pooled	-8.8131	-11.8452	-5.7811	12.6825	11.7552	13.7699
Diff (1-2)	Satterthwaite	-8.8131	-11.4230	-6.2032			

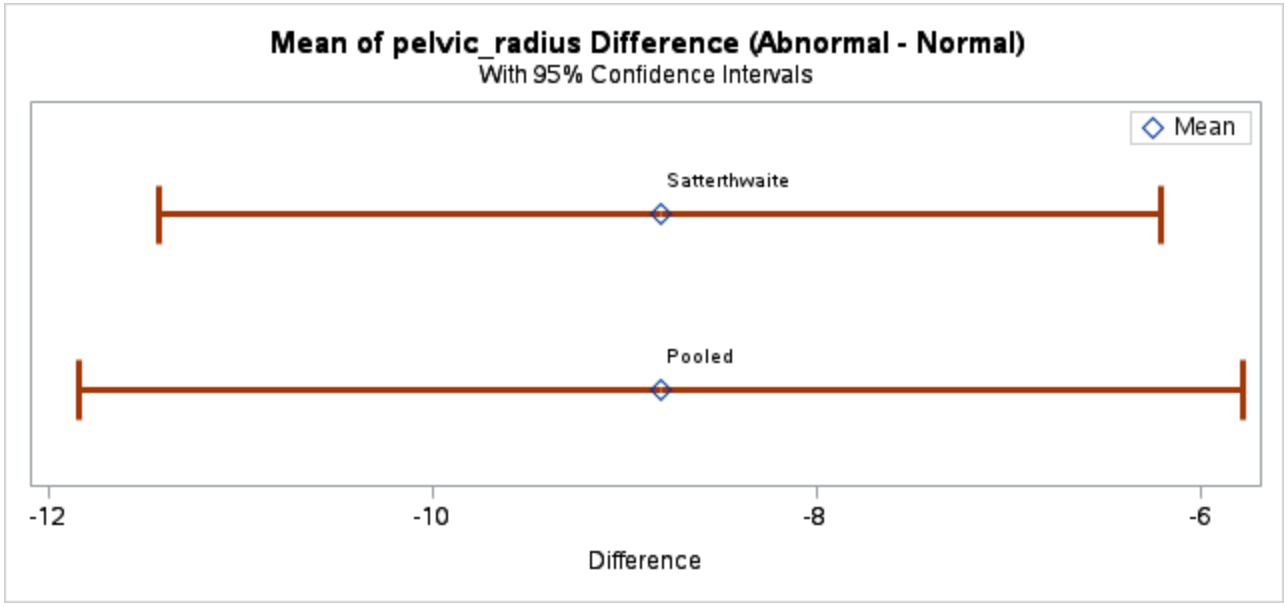
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	308	-5.72	<.0001
Satterthwaite	Unequal	282.35	-6.65	<.0001

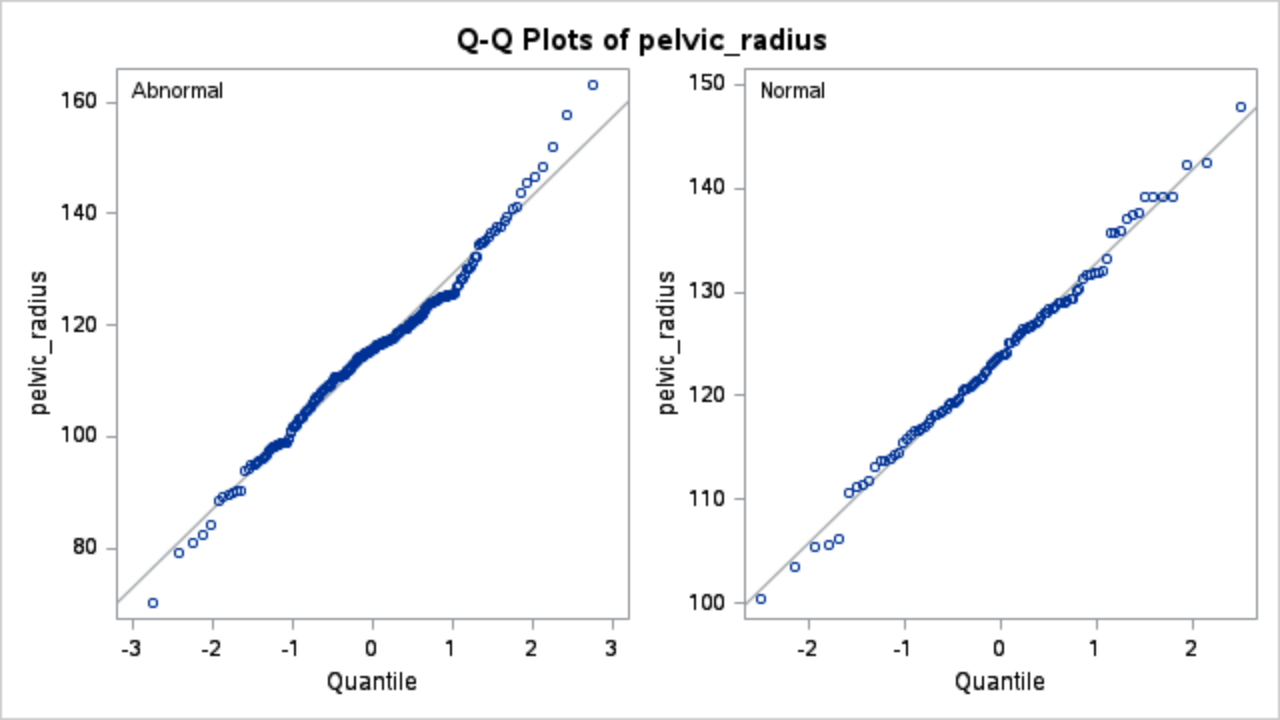
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	209	99	2.44	<.0001


Distribution of pelvic_radius
Abnormal
Percent
Normal
Class_att
Abnormal
Normal
75
100
125
150
175
pelvic_radius
Normal
Kernel


Distribution of pelvic_radius
Abnormal
Percent
Normal
75
100
125
150
175
pelvic_radius
Normal
Kernel


Distribution of pelvic_radius
Class_att
Abnormal
Normal
168
163
174 156
76
181
84
86
225
305
80
100
120
140
160
pelvic_radius


Mean of pelvic_radius Difference (Abnormal - Normal)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
-12
-10
-8
-6
Difference



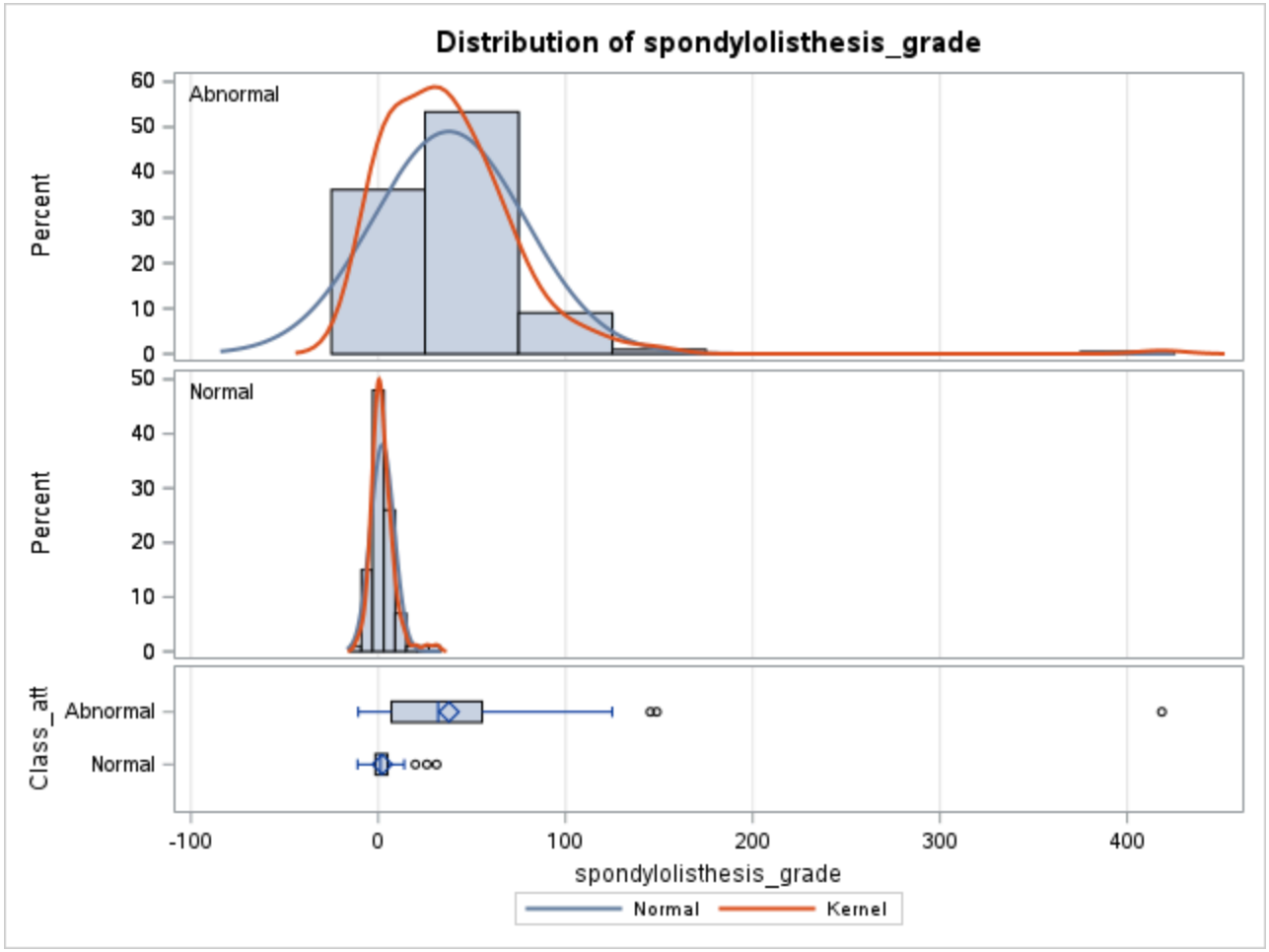
Variable: spondylolisthesis_grade

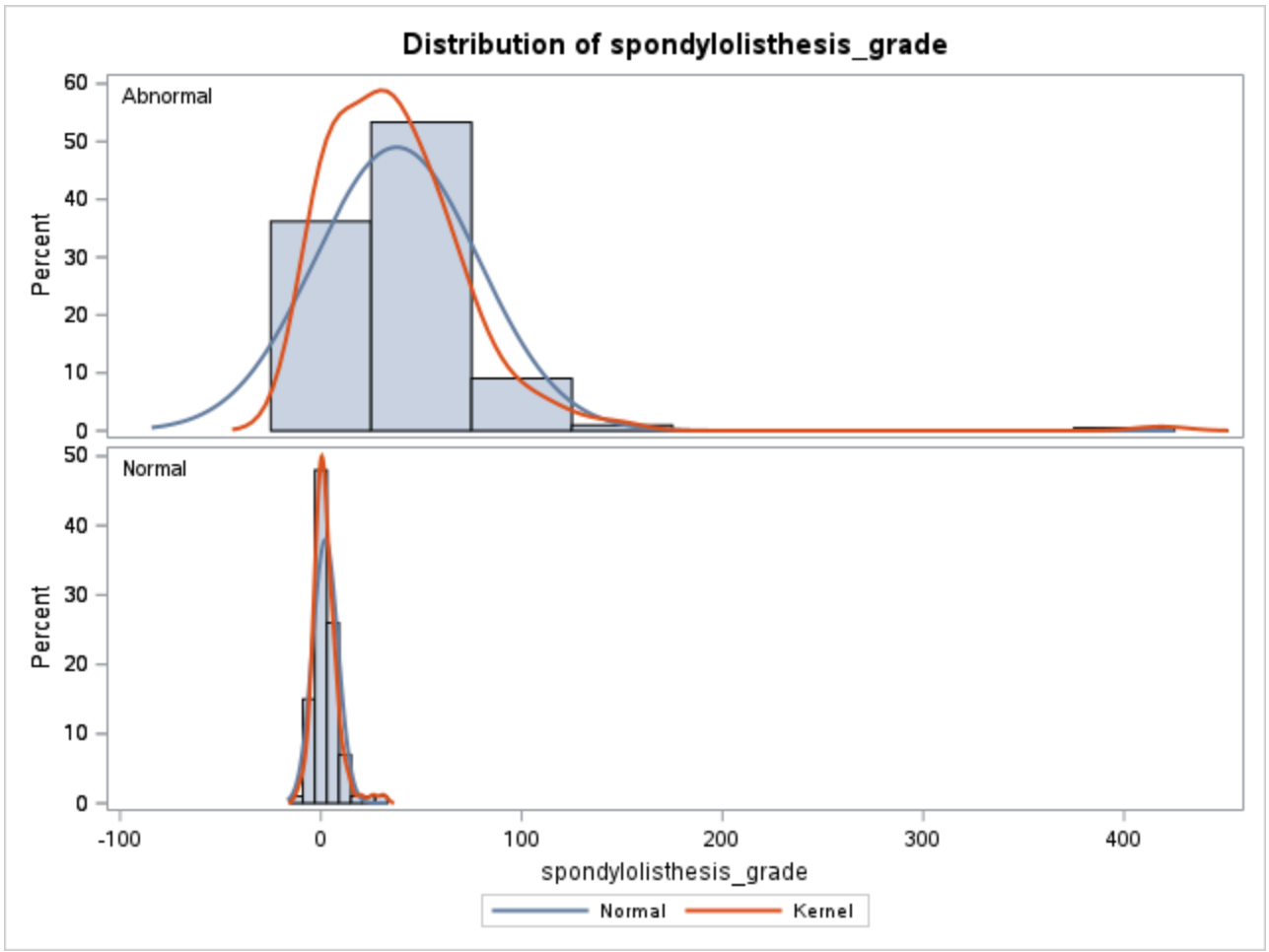
Class_att	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Abnormal		210	37.7777	40.6967	2.8083	-10.6759	418.5
Normal		100	2.1866	6.3075	0.6307	-11.0582	31.1728
Diff (1-2)	Pooled		35.5911	33.7143	4.0962		
Diff (1-2)	Satterthwaite		35.5911		2.8783		

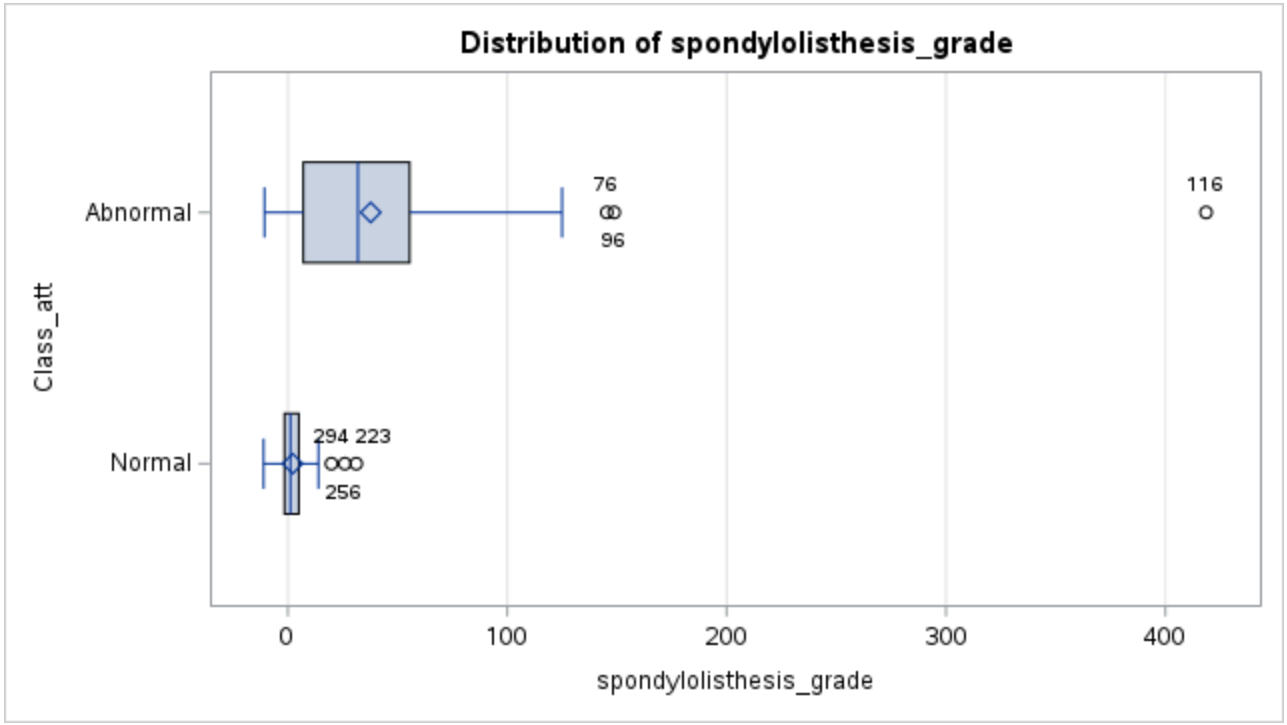
Class_att	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Abnormal		37.7777	32.2414	43.3140	40.6967	37.1412	45.0110
Normal		2.1866	0.9350	3.4381	6.3075	5.5380	7.3272
Diff (1-2)	Pooled	35.5911	27.5310	43.6513	33.7143	31.2492	36.6049
Diff (1-2)	Satterthwaite	35.5911	29.9198	41.2624			

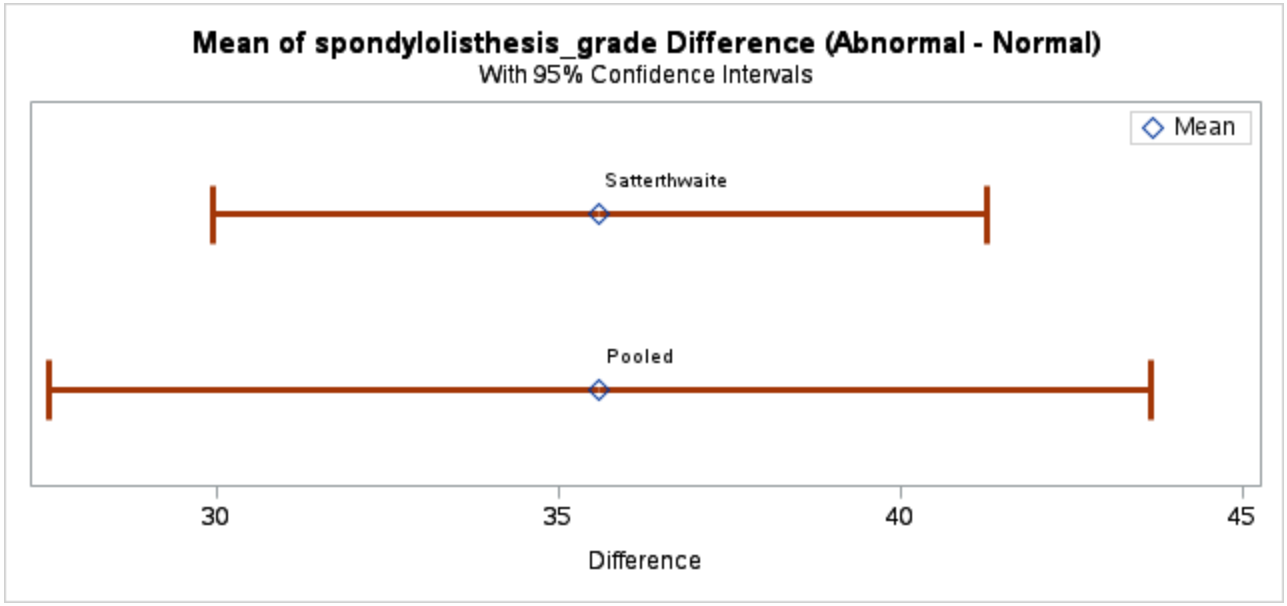
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	308	8.69	<.0001
Satterthwaite	Unequal	229.39	12.37	<.0001

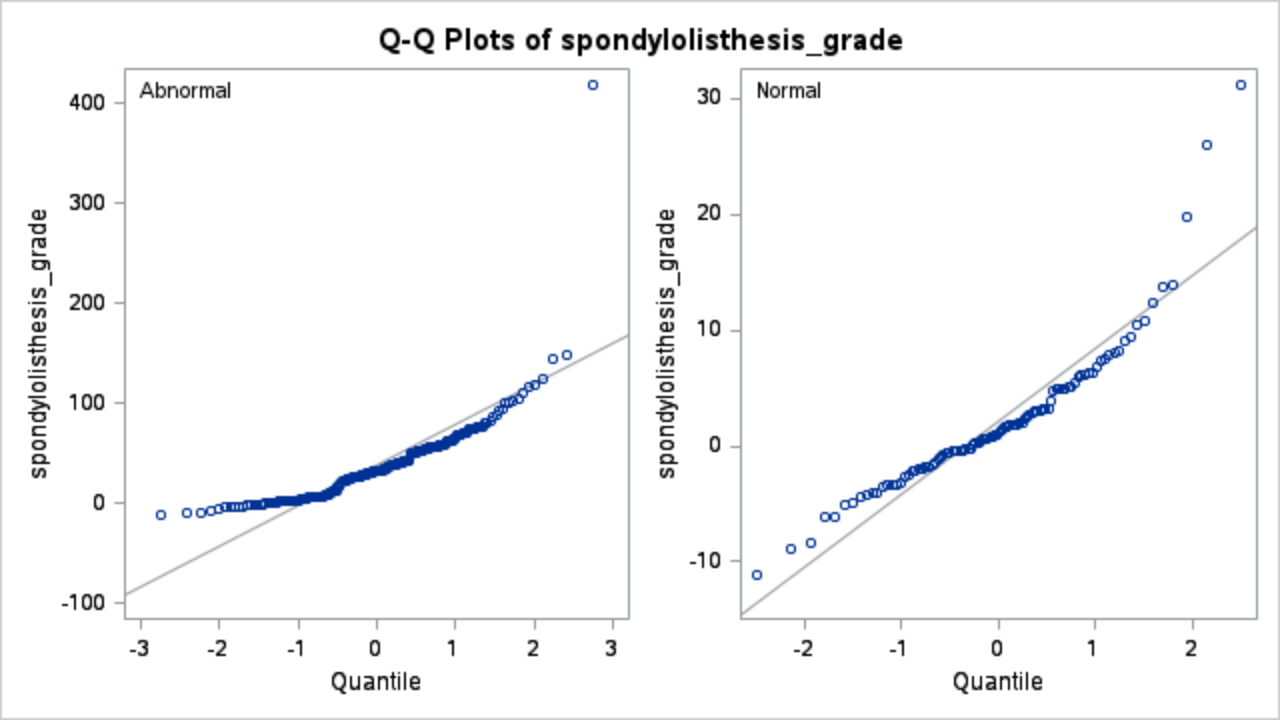
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	209	99	41.63	<.0001


Distribution of spondylolisthesis_grade
Abnormal
Normal
Percent
Class_att
Abnormal
Normal
-100
0
100
200
300
400
spondylolisthesis_grade
Normal
Kernel


Distribution of spondylolisthesis_grade
Abnormal
Normal
Percent
-100
0
100
200
300
400
spondylolisthesis_grade
Normal
Kernel


Distribution of spondylolisthesis_grade
Class_att
Abnormal
Normal
76
96
116
294 223
256
0
100
200
300
400
spondylolisthesis_grade


Mean of spondylolisthesis_grade Difference (Abnormal - Normal)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
30
35
40
45
Difference



The LOGISTIC Procedure

Model Information	
Data Set	SAS.SPINE
Response Variable	Class_att
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	310
Number of Observations Used	310

Response Profile		
Ordered Value	Class_att	Total Frequency
1	Abnormal	210
2	Normal	100

Probability modeled is Class_att='Abnormal'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	391.856	190.261
SC	395.592	212.680
-2 Log L	389.856	178.261

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq

Testing Global Null Hypothesis: BETA=0			
Test	**Chi-Square**	**DF**	**Pr > ChiSq**
Likelihood Ratio	211.5947	5	<.0001
Score	105.7904	5	<.0001
Wald	62.9991	5	<.0001

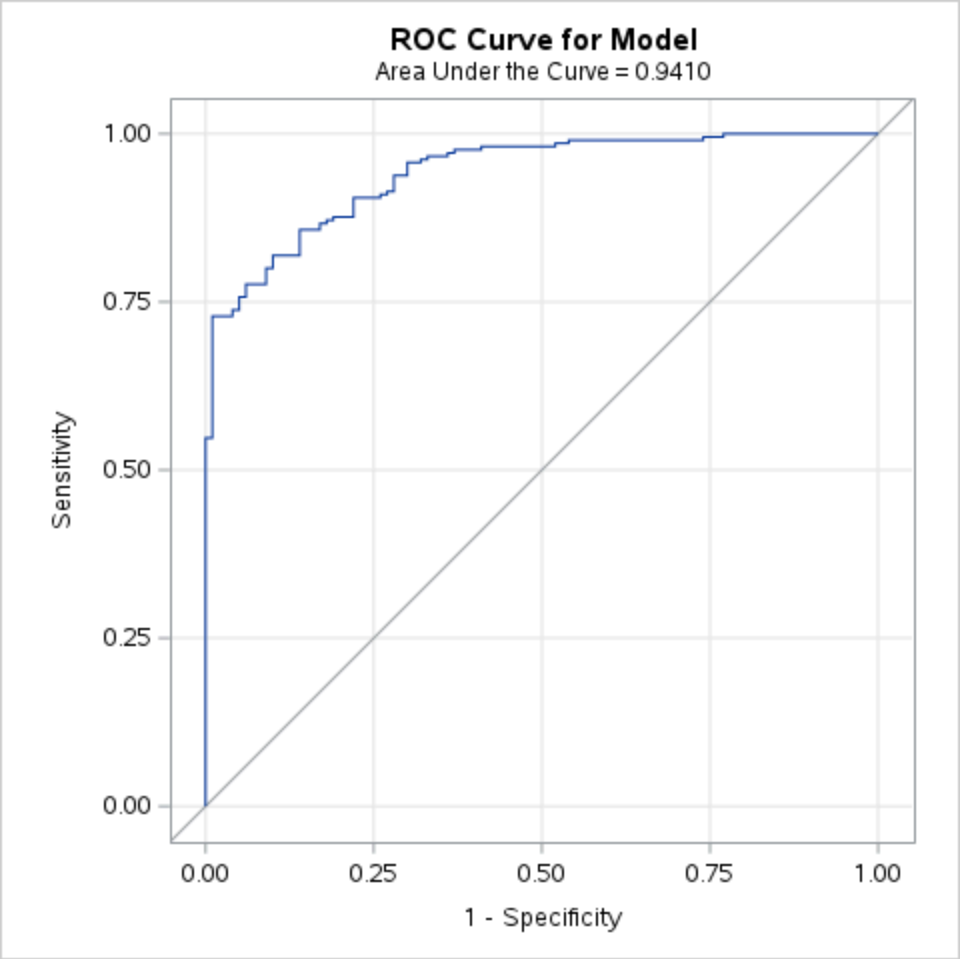
Note: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

sacral_slope =	pelvic_incidence - pelvic_tilt

Analysis of Maximum Likelihood Estimates					
Parameter	**DF**	**Estimate**	**Standard Error**	**Wald Chi-Square**	**Pr > ChiSq**
Intercept	1	15.1692	3.2793	21.3978	<.0001
pelvic_incidence	1	-0.0968	0.0312	9.6061	0.0019
pelvic_tilt	1	0.1724	0.0390	19.5061	<.0001
lumbar_lordosis	1	-0.0187	0.0228	0.6707	0.4128
sacral_slope	0	0	.	.	.
pelvic_radius	1	-0.1069	0.0229	21.7105	<.0001
spondylolisthesis_gr	1	0.1682	0.0232	52.5565	<.0001

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	94.1	**Somers' D**	0.882
Percent Discordant	5.9	**Gamma**	0.882
Percent Tied	0.0	**Tau-a**	0.387
Pairs	21000	**c**	0.941

Odds Ratio Estimates and Wald Confidence Intervals				
Effect	**Unit**	**Estimate**	**95% Confidence Limits**	
pelvic_incidence	1.0000	0.908	0.854	0.965
pelvic_tilt	1.0000	1.188	1.101	1.283
lumbar_lordosis	1.0000	0.981	0.939	1.026
pelvic_radius	1.0000	0.899	0.859	0.940
spondylolisthesis_gr	1.0000	1.183	1.131	1.238



The DISCRIM Procedure

Total Sample Size	310	**DF Total**	309
Variables	6	**DF Within Classes**	308
Classes	2	**DF Between Classes**	1

Number of Observations Read	310
Number of Observations Used	310

Class Level Information

Class_att	Variable Name	Frequency	Weight	Proportion	Prior Probability
Abnormal	Abnormal	210	210.0000	0.677419	0.500000
Normal	Normal	100	100.0000	0.322581	0.500000

Pooled Covariance Matrix Information

Covariance Matrix Rank	Natural Log of the Determinant of the Covariance Matrix
5	14.22388

The DISCRIM Procedure

Generalized Squared Distance to Class_att

From Class_att	Abnormal	Normal

Generalized Squared Distance to Class_att		
From Class_att	Abnormal	Normal
Abnormal	0	2.35539
Normal	2.35539	0

Linear Discriminant Function for Class_att		
Variable	Abnormal	Normal
Constant	-74.13204	-84.69953
pelvic_incidence	0.31481	0.32584
pelvic_tilt	-0.03166	-0.08120
lumbar_lordosis	-0.09340	-0.11771
sacral_slope	0.53686	0.58252
pelvic_radius	1.01207	1.10228
spondylolisthesis_grade	-0.18065	-0.21810

The DISCRIM Procedure
Classification Summary for Calibration Data: SAS.SPINE
Resubstitution Summary using Linear Discriminant Function

Number of Observations and Percent Classified into Class_att			
From Class_att	Abnormal	Normal	Total
Abnormal	160 76.19	50 23.81	210 100.00
Normal	10 10.00	90 90.00	100 100.00
Total	170 54.84	140 45.16	310 100.00
Priors	0.5	0.5	

Error Count Estimates for Class_att			
	Abnormal	Normal	Total
Rate	0.2381	0.1000	0.1690
Priors	0.5000	0.5000	

The DISCRIM Procedure
Classification Summary for Calibration Data: SAS.SPINE
Cross-validation Summary using Linear Discriminant Function

Number of Observations and Percent Classified into Class_att			
From Class_att	Abnormal	Normal	Total
Abnormal	156 74.29	54 25.71	210 100.00
Normal	10 10.00	90 90.00	100 100.00
Total	166 53.55	144 46.45	310 100.00
Priors	0.5	0.5	

Error Count Estimates for Class_att			
	Abnormal	Normal	Total
Rate	0.2571	0.1000	0.1786

Error Count Estimates for Class_att			
	Abnormal	Normal	Total
Priors	0.5000	0.5000	